Allamaprabhu Ani

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Research Summary

PhD student and lead developer of PhAST, an open, matrix-free, PyTorch-native phase-field fracture solver. Research covers GPU finite-element kernels, fracture benchmarks, and autograd-compatible scalar inverse recovery; ongoing thesis work examines spatial inverse problems and solver-audited learning. Lead author of the published PhAST paper and *Engineering Fracture Mechanics* review (Vol. 332, Art. 111778), with collaborators at EPFL, the University of Florida, and City St George's.

Education

PhD, Mechanical Engineering *Jul 2023 – present*
City St George's, University of London, School of Science and Technology
Writing-up status.
Research: differentiable fracture simulation, inverse analysis, and hybrid physics–ML methods.
Supervisor: Professor Asa Barber. Former supervisor: Dr Sathiskumar A. Ponnusami FRAeS.

MTech, Design and Manufacturing *Sep 2021 – May 2023*
National Institute of Technology Silchar, CGPA 9.88/10. Best Student Award.
Thesis: Machine learning for fatigue life prediction in aluminium alloys. Advisor: Dr A.B. Deoghare.

BE, Mechanical Engineering *Aug 2016 – Jul 2020*
GM Institute of Technology, Davangere (Visvesvaraya Technological University), CGPA 8.47/10.

Awards and Professional Recognition

- **Yeoman and Travelling Scholarship, Worshipful Company of Tin Plate Workers** (Mar 2026), awarded for "AI-accelerated modelling for fracture prediction" and used to support presentation at WCCM-ECCOMAS 2026.
- **Dean's Award for Outstanding Teaching Support**, School of Science and Technology, City St George's, University of London (Apr 2026). Nominated by students.
- **Associate Fellow of the Higher Education Academy (AFHEA)**, Advance HE (Oct 2024; credential PR301728).
- **Member, Royal Aeronautical Society** (2026).
- **Funded PhD Studentship** "Modelling for Failure Analysis," City St George's, University of London (2023).
- **Best Student Award**, MTech Design and Manufacturing, National Institute of Technology Silchar (2023).
- **Inspiring Student Award for Mechanical Engineering**, Visvesvaraya Technological University (Dec 2020).

Publications

Journal articles

J1. **Ani, A.**, Molinari, J.-F., Subhash, G., Ponnusami, S.A. (2027). PhAST: A differentiable, matrix-free finite-element solver for dynamic phase-field fracture: Formulation, benchmarks, and inverse analysis. *Computer Methods in Applied Mechanics and Engineering*, 464, 119547. doi:10.1016/j.cma.2026.119547.

J2. **Ani, A.S.**, Nakka, R., Subhash, G., Molinari, J.-F., Ponnusami, S.A. (2026). Machine learning for computational fracture and damage mechanics: Status and perspectives. *Engineering Fracture Mechanics*, 332, 111778. doi:10.1016/j.engfracmech.2025.111778. *Open access; field-level review paper.*

J3. Srinivasa, C.V., **Ani, A.S.**, Jyothi Prasad, B.M. (2020). Protective Coatings for Bio-Composites: A Review. *IOP Conference Series: Materials Science and Engineering*, 925, 012048. doi:10.1088/1757-899X/925/1/012048.

Book chapters and conference papers

C1. **Ani, A.S.**, Deoghare, A.B. (2024). Leveraging Machine Learning for Enhanced Fatigue Life Prediction in Aluminium Alloys. In: *Lecture Notes in Mechanical Engineering*, Springer. doi:10.1007/978-981-97-7535-4_61.

In preparation

- Manuscripts on spatial inverse recovery and machine-learning-assisted fracture simulation, arising from ongoing thesis work.

Research Experience

PhD Student, Computational Mechanics and Machine Learning *Jul 2023 – present*
City St George's, University of London

- Lead developer and maintainer of PhAST, a matrix-free, PyTorch-native, GPU-accelerated phase-field fracture solver with autograd-compatible analysis.
- Demonstrated scalar fracture-toughness recovery with gradient checks in the published article; current thesis work evaluates spatial parameter recovery, inclusion localisation, and solver-audited learned assistance.
- Documented six quasi-static and dynamic benchmark configurations, with reference comparisons described in the published article.
- Maintained reproducible HPC workflow on Hyperion2 (Slurm) with local Mac CPU and WSL CUDA development environments.

Visiting Researcher, LSMS Lab, EPFL *2025*
Host: *Prof. Jean-François Molinari* (École Polytechnique Fédérale de Lausanne).

- Collaborated on the 2026 *Engineering Fracture Mechanics* review and differentiable methods for solid mechanics; the visit informed the autograd-through-staggered-solver framework in the PhD code base.

MTech Researcher, National Institute of Technology Silchar *2021 – 2023*
Advisor: *Dr A.B. Deoghare*. Machine-learning surrogate models for fatigue-life prediction across aluminium alloy compositions (scikit-learn, MATLAB). Best Student Award.

Founder and Tool Developer, Aeroknacks India Ltd *2020 – 2021*

- Built a library of validated aerospace hand-calculation automation tools and Excel-VBA workflows based on E.F. Bruhn, Michael Niu, Boeing Design Manuals, and ESDU references.
- Implemented Cozzone f_0 plastic bending, lug strength, fastener load transfer, column buckling, modal analysis, ABD, Neuber non-linear stress, and a NASTRAN `.f06` parser. Tools shipped to an aerospace engineering firm.

Research Intern, NMCAD Lab, Indian Institute of Science Bengaluru
Research Intern, CGPL Lab, Indian Institute of Science Bengaluru (Feb 2018): Pressure Swing Adsorption for 99.997%-purity hydrogen from biomass gasifier.

Teaching and Outreach

Teaching, City St George's (2023 - present)

- Graduate Teaching Assistant, EG1004 Introduction to Mechanics, Materials and Manufacturing.
- Graduate Teaching Assistant, EG1002 Wind Turbine Challenge: Tower Design, and EG1002 Siemens NX CAD teaching support.
- Created and maintain open PyTorch and scientific-machine-learning tutorials while assisting Dr Sathiskumar A. Ponnusami's teaching at Queen Mary University of London.

Teaching, National Institute of Technology Silchar (2021 - 2023)

- Teaching Assistant: ME-316 Machine Design Laboratory (SolidWorks, ANSYS, FEA); ME-202 Theory of Machines.

Workshop co-organisation

- Co-organiser, EPSRC-supported M2L (Mechanics and Machine Learning) Workshop, City St George's (July 2024 and July 2025), with speakers from Imperial, EPFL, the University of Florida, TU Delft, IIT Delhi, and City St George's.
- Managed venue booking, Qualtrics registration, event posters and communications, speaker presentation support, guest reception, and Teams broadcasting for an audience of around 70 researchers attending online and in person.

Selected Talks and Teaching Events

- **WCCM–ECCOMAS 2026, Munich** (July 2026): presented work on hybrid neural-operator and mechanics-solver workflows for fracture simulation.
- **UKACM Autumn School** (September 2026): assisted Dr Sathiskumar Anusuya Ponnusami with interactive mechanics and machine-learning teaching, introducing PhAST and its use cases. Course material.

Open-Source Software and Online Presence

- **PhAST**: matrix-free, differentiable PyTorch software for phase-field fracture (source; documentation).
- `geometric-transformations`: 2D/3D affine transforms with MATLAB Live Scripts (source).
- **Personal site**: allamaprabhuani.github.io, research notes and technical tutorials.

Technical Skills

Scientific computing and ML: Python, PyTorch (autograd, CUDA), NumPy, SciPy, scikit-learn, differentiable physics, inverse problems, neural operators, and physics-informed neural networks.
Mechanics and engineering: phase-field fracture, finite elements, Akantu, FEniCSx, COMSOL, MSC Nastran/Patran, Abaqus, ANSYS, MATLAB, and Excel-VBA.
Research infrastructure: Slurm, Linux, Git, Sphinx, GitHub Actions, Python packaging, LaTeX, and Bash.